Jhih-Jhong (Johnny) Lin

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Education

Bachelor of Science in Computer Science and Information Engineering Taipei, Taiwan
National Taiwan University (NTU) *Aug. 2022 - Jul. 2026 (Expected)*

Cumulative GPA: 4.24/4.30 | **Cumulative Class Rank**: 8/196 (as of 2022–2025)
Admitted to: M.S. Program in Computer Science and Information Engineering, NTU (Fall 2026)
Relevant Courses: Algorithm Design and Analysis (A+), Quantum Algorithms (A+), Cryptography and Network Security (A+), Post-quantum Cryptography (A-), Programming Language Theories and Type Systems (A+), Compiler Design (A+), Intuitionistic Logic (A+), Formal Methods (A), Formal Languages and Automata Theory (A+), Intro to Abstract Algebra I (A-), Operating Systems (A+), Virtual Machines (A), Computer Networks (A+)

Research Experience

Research Intern / Project Assistant Taipei, Taiwan
Institute of Information Science (IIS), Academia Sinica | Advisor: Prof. ***Bow-Yaw Wang*** *Jul. 2025 - Present*

- Developing abstract interpretation techniques for verification of floating-point cryptographic programs.

Undergerduate Researcher Taipei, Taiwan
Network Security Lab (NSLab), CSIE, National Taiwan University | Advisor: Prof. ***Hsu-Chun Hsiao*** *Sep. 2024 - Present*

- Investigated integration of directed fuzzers with recurring vulnerability detection tools.
- Currently exploring formal methods, including static analysis and cryptographic protocol verification.

Technical Experience

Netowrk Administrator (Member of NASA Team) Taipei, Taiwan
Department of Computer Science and Information Engineering, National Taiwan University *Sep. 2023 - Jun. 2025*

- Maintained switch infrastructure serving 600+ faculty members and students.
- Built automated monitoring and backup workflows for newly deployed switches.

Teaching Experience

Teaching Assistant, Introduction to Computational Logic Taipei, Taiwan
Department of Computer Science and Information Engineering, National Taiwan University *Fall 2025*

- Evaluated biweekly homework assignments for a class of 20+ students.

Teaching Assistant, Network Administration and System Administration Taipei, Taiwan
Department of Computer Science and Information Engineering, National Taiwan University *Spring 2025*

- Co-designed homework and exam problems for a class of 50+ students, providing hands-on experience with shell-scripting and computer security concepts.

Selected Projects

MerkleDir.sh: s shell script that creates Merkle trees from directories and generates/verifies inclusion proofs
Code (Bash) | Developed as a TA for Network Administration and System Administration

CTF Challenge Design: Cryptography, Fuzzing, and DNS Cache Poisoning
DNS Cache Poisoning | Cryptography and Fuzzing | Developed as a TA for Network Administration and System Administration

An Evaluation on Possibilities of RCE and Defense Mechanism in Electron-Based Applications
Co-authored with: Gang-Yu Fan, Tsan-Huan Hsu, Yi-Ting Lo
Project Report | Cryptography and Network Security Course Project

Grey-Box Symbolic Execution with Likely-Invariant Generation in CPAchecker
Code (Java) | Project Report | Formal Methods Course Project

Terminal-Based P2P and Centralized Chatroom Implementation
Co-authored with: Wen-Chun Kao
Code (C++) | Computer Networks Course Project

Awards

Dean's List, NTU, Fall 2023 *Mar. 2024*

Dean's List, NTU, Fall 2022 *Mar. 2023*

Skills and Interests

Languages: Mandarin (native); English (advanced reading & writing, intermediate speaking; TOEFL iBT 103/120)

Programming: C, Python, Ocaml, Bash; working knowledge of C++, Java, and Haskell

Software: Linux, Git, LaTeX, Docker, CPAchecker, CryptoLine, Compiler (flex, yacc), Fuzzing (AFL++)

Research Interests: Formal verification, program analysis, cryptography

Games: Arcaea